JYX
JYVÄSKYLÄN YLIOPISTO
UNIVERSITY OF JYVÄSKYLÄ

This is a self-archived version of an original article. This version may differ from the original in pagination and typographic details.

Author(s): Poudel, Dinesh; Shahzad, Faisal

Title: The Failed Helix: Understanding Why University–Stakeholder Collaborations Collapse

Year: 2026

Version: Published version

Copyright: © 2026 the Authors

Rights: CC BY 4.0

Rights url: https://creativecommons.org/licenses/by/4.0/

Please cite the original version:

Poudel, D., & Shahzad, F. (2026). The Failed Helix: Understanding Why University–Stakeholder Collaborations Collapse. Triple Helix, Early online. https://doi.org/10.1163/21971927-bja10079

The University of Jyväskylä's Rights Retention Strategy (RRS) enables accepted versions (final drafts) to be made openly available through self-archiving. For more information, see: https://www.jyu.fi/en/rrs

BRILL

TRIPLE HELIX (2026) 1–36

TRIPLE HELIX
brill.com/thj

The Failed Helix: Understanding Why University–Stakeholder Collaborations Collapse

Dinesh Poudel | ORCID: 0000-0001-8674-282X
Department of Management and Leadership, University of Jyväskylä, Jyväskylä, Finland
School of Business, Design and Technology, Häme University of Applied Sciences, Valkeakoski, Finland
Corresponding author
dinesh.poudel@hamk.fi, dpoudext@jyu.fi

Faisal Shahzad | ORCID: 0000-0001-8457-7322
School of Business, Design and Technology, Häme University of Applied Sciences, Valkeakoski, Finland
faisal.shahzad@hamk.fi

Received 27 February 2026 | Accepted 1 September 2026 | Published online 13 October 2026

Abstract

The Triple Helix and Quadruple Helix frameworks have become dominant paradigms for understanding university–industry–government collaboration in regional innovation systems. The empirical literature focuses on successful collaborations while neglecting those that fail, dissolve, or never achieve operational status. This paper investigates why university–stakeholder collaborations fail and how breakdown unfolds. Drawing on institutional theory and inter-organisational collaboration research, we develop a framework distinguishing five failure forms: dissolution, zombie collaboration, stillbirth, fragmentation, and mission drift, mapped across three lifecycle stages. Through case studies of three failed collaborations in Finland, we demonstrate that structural vulnerability and institutional misalignment operate as latent conditions, while environmental shocks and relational erosion function as proximate triggers at critical junctures. We show that civil society inclusion in Quadruple Helix arrangements does not inherently produce equitable participation, representing

Published with license by Koninklijke Brill BV | DOI:10.1163/21971927-BJA10079
© DINESH POUDEL AND FAISAL SHAHZAD, 2026 | ISSN: 2197-1927 (online)
This is an open access article distributed under the terms of the CC BY 4.0 license.

a failure dynamic Triple Helix framework cannot anticipate. These findings advance theoretical understanding and offer practical implications for designing more resilient partnerships.

Keywords

Triple Helix – Quadruple Helix – collaboration failure – failure typology – zombie collaboration – university–industry–government – institutional logics – Finland

1 Introduction

University–stakeholder collaboration has become a cornerstone of regional innovation policy worldwide. Governments invest substantial resources in fostering partnerships between higher education institutions, industry, and public agencies, premised on the belief that such collaborations drive economic development, knowledge transfer, and societal impact. The Triple Helix model, introduced by Etzkowitz and Leydesdorff (1995, 2000), has provided the dominant theoretical framework for understanding these dynamics and has subsequently been extended to Quadruple and Quintuple Helix formulations that incorporate civil society and ecological considerations (Carayannis & Campbell, 2009, 2010).

Yet despite decades of research and policy attention, we know remarkably little about why such collaborations fail. Empirical literature is characterised by what might be termed a survivorship bias: studies mostly examine collaborations that exist and function, while those that have dissolved, stagnated, or never achieved operational status remain largely invisible to scholarly inquiry. This creates a distorted picture in which success stories dominate, and the conditions for failure remain poorly understood.

Understanding only success limits our ability to identify the necessary and sufficient conditions for effective collaboration. Factors that appear in successful cases may be merely incidental rather than causal if they also appear in failed cases. A research design that relies solely on existing collaborations cannot, by definition, tell us what distinguishes sustainable partnerships from fragile ones. Similarly, practitioners launching new collaboration initiatives lack guidance on pitfalls to avoid and warning signs to monitor. The result is that similar mistakes may recur across contexts, wasting resources and eroding stakeholder trust in collaborative approaches.

We address this gap by examining the question: *Why do some university–stakeholder collaborations fail, and what conditions and mechanisms lead to their dissolution?* We adopt a qualitative multiple-case study approach to examine three collaborations that collapsed, became inactive, or became significantly fragmented in the Finnish context. Finland's mature innovation economy and active promotion of Helix-style partnerships over two decades (Cai & Lattu, 2022) make the failure of such collaborations analytically revealing, pointing to endogenous mechanisms of breakdown rather than mere resource deficiency. The Finnish research context is described in detail in Section 1.1 below.

1.1 *Research Context: University-Stakeholder Collaboration in Finland*

Finland provides an empirically appropriate context for this study. As a small, open economy highly dependent on knowledge-intensive industries, Finland has invested substantially in building collaborative innovation infrastructure over three decades (European Commission, Policy Support, 2025; Research Council Finland, 2025). The national innovation system is characterised by strong universities, proactive government agencies, and a policy culture that has actively promoted university–industry–government collaboration since the 1990s (Cai & Lattu, 2022). Finnish universities operate under a dual mandate of academic excellence and societal engagement. This creates institutional pressures that both encourage collaboration and generate the internal tensions our findings illuminate.

EU structural funds have been a major driver of university–stakeholder collaborations, particularly in regional development contexts. Multi-year EU-funded programmes have financed innovation hubs, joint research centres, and industry–academic partnerships across Finnish regions. However, the project-based, time-limited nature of EU funding creates a structural dependency: collaborations often develop during funded periods without establishing sustainable governance or resource models, leaving them acutely vulnerable when funding cycles end. This funding architecture shapes the failure dynamics we observe across our three cases.

Finnish higher education underwent significant structural reforms in the 2010s (Ampuja & Horowitz, 2025), including university mergers and the introduction of performance-based funding tied to research output and degree completions. These reforms strengthened publication-based incentive structures within universities, intensifying the tension between academic career rewards and collaboration engagement that our findings on institutional misalignment documents. At the same time, regional development agencies – key partners in our cases – operate under political accountability pressures that

require visible, short-term outputs, creating a fundamental temporal mismatch with academic research timelines.

Against this backdrop, the three collaborations we studied – Projects A, B, and C – were initiated under conditions broadly supportive of Helix-style partnerships. Their failure, therefore, cannot be attributed to an unsupportive policy environment; rather, it points to endogenous mechanisms of breakdown that our analysis seeks to illuminate.

Our contribution is fourfold. First, we develop and empirically apply a typology of collaboration failure that distinguishes five forms of breakdown: 1) *dissolution*, 2) *zombie collaboration*, 3) *stillbirth*, 4) *fragmentation*, and 5) *mission drift*. Each of these reflects a distinct configuration of contributing factors. Second, we identify and organise the factors that contribute to failure across *relational, structural, institutional, and environmental dimensions*, showing, through cross-case analysis, which factors appear consistently and which are case-specific. Third, we examine the dynamics of failure, revealing how latent vulnerabilities interact with environmental triggers to produce cascading breakdown: a process that is often invisible to participants until recovery becomes unlikely. Fourth, we position these findings explicitly within Helix theory, showing how the presence of university, industry, and government actors is necessary but not sufficient, and that genuine Quadruple Helix principles (particularly equitable partner inclusion) may be a protective factor against failure. In doing so, we complement the existing literature on collaboration success and offer practical implications for designing more resilient partnerships.

Underlying these contributions, we make a single theoretical argument: that failure in university–stakeholder collaborations is a process, not an event, driven by the interaction of latent structural conditions and proximate environmental triggers rather than by any single dominant cause. This shifts the analytical focus in Helix research from asking whether collaborations include the right actors to asking whether they have built the governance, relational, and institutional conditions needed to survive over time.

2 Theoretical Background

The Triple Helix model has become a foundational framework for regional innovation policy over the past three decades. Governments across Europe and beyond have drawn on its logic to design and fund university–industry–government partnerships as instruments of regional economic development, knowledge transfer, and societal impact (Etzkowitz & Leydesdorff, 2000; Carayannis & Campbell, 2009). In Finland, as in many advanced

innovation economies, Helix thinking has directly shaped the architecture of publicly funded collaboration programmes, science parks, innovation hubs, and regional development initiatives. The widespread adoption of the Helix framework makes understanding its limitations and particularly the conditions under which Helix-inspired collaborations fail, both theoretically important and practically urgent. The following sections review the Helix literature, the inter-organisational collaboration literature, and the organisational failure literature to develop the conceptual foundations for our analysis.

2.1 *University–Stakeholder Collaboration in Helix Models*

The Triple Helix model conceptualises innovation as emerging from dynamic interactions among three institutional spheres: university, industry, and government (Etzkowitz & Leydesdorff, 1995, 2000). Unlike earlier linear models of innovation, the Triple Helix emphasises recursive relationships, hybrid organisations, and the idea that each sphere takes on new roles. Universities become entrepreneurial, engaging in technology transfer and spin-off creation; firms conduct basic research; governments act as venture capitalists and innovation brokers. The overlapping spaces between helices generate innovation through knowledge exchange, resource combination, and institutional adaptation.

Carayannis and Campbell (2009, 2010) extended this framework to incorporate additional helices. The Quadruple Helix adds civil society and the media-based public as a fourth sphere, emphasising that advanced innovation systems require democratic participation and knowledge democracy. The Quintuple Helix further incorporates the natural environment, positioning ecological challenges as both constraints on and drivers of innovation. Carayannis et al. (2022) argue that these extensions represent a fundamentally different conceptualisation: genuine Quadruple and Quintuple Helix innovation systems require democratic governance in substance, not merely in form. This normative dimension has important implications for the design of collaboration. The partnerships that formally include multiple actors but marginalise some from meaningful decision-making may replicate the governance failures of earlier, less inclusive models.

Despite theoretical development, the Helix literature exhibits notable limitations. It predominantly studies collaborations that exist and function through case studies of innovation hubs, science parks, or regional partnerships, while failed or dissolved collaborations rarely feature. The few exceptions are instructive: Grey et al. (2011) examined failures in industry–university cooperative research centres, identifying governance deficits and misaligned partner expectations as recurring causes, while Thomasson et al. (2021) analysed accountability failures in Triple Helix initiatives, revealing how public interest considerations are frequently marginalised in practice. Yet these

studies remain isolated contributions rather than the foundation of a systematic research programme. This creates a selection bias that may overstate the effectiveness of Helix arrangements and obscure the conditions necessary for their success. Literature also tends toward normative prescription, implicitly assuming that more collaboration is better, rather than critically examining when collaboration works and when it does not. Finally, relatively little attention has been paid to collaboration dynamics over time, including the processes through which partnerships form, evolve, and potentially dissolve (Etzkowitz & Leydesdorff, 2000; Cai & Lattu, 2022).

Cai and Lattu (2022) provide a systematic comparison of Triple and Quadruple Helix models, noting that current empirical research has not clearly delineated the two models, generating ambiguity about which framework to apply in practice. Empirical gaps compound this theoretical ambiguity: understanding when and why Helix collaborations fail remains a significant lacuna. Our study directly addresses this gap by bringing failure into the centre of Helix empirical inquiry.

2.2 *Collaborative Inertia and Beyond: What the Inter-Organisational Literature Tells Us*

While the Helix literature has paid limited attention to failure, the broader inter-organisational relations literature offers important, if underutilised, theoretical resources. In this study, collaboration failure is defined as the inability of a university–stakeholder partnership to sustain the conditions necessary for continued meaningful joint activity, irrespective of whether formal dissolution occurs. This encompasses partnerships that formally ended, those that became operationally inactive while maintaining nominal existence, and those that significantly deviated from their founding objectives. Failure is understood as a process rather than an event, one that is socially constructed, retrospectively interpreted, and shaped by the interaction of relational, structural, institutional, and environmental conditions over the collaboration lifecycle. Huxham and Vangen (2005) use the term 'collaborative inertia' to describe situations in which partnership produces little or nothing despite its formal existence, contrasting it with 'collaborative advantage' in which genuine synergies emerge. Their work identifies numerous structural and relational impediments: power imbalances, conflicting goals, communication breakdowns, and the difficulty of coordinating across organisational boundaries, where different cultures, timelines, and accountability structures are present. Where our study extends their contribution is in examining not merely inertia but the full process of breakdown (from early warning signs through critical junctures to dissolution) and in situating this process within the specific institutional context of Helix collaborations.

Ring and Van de Ven (1994) proposed a developmental model of inter-organisational relationships progressing through negotiation, commitment, and execution. Their framework highlights how relationships can break down at any stage: negotiations may fail to produce agreement; commitments may not be honoured; execution may reveal incompatibilities that undermine the partnership. Ariño and de la Torre (1998) extended this work by examining learning and adaptation in alliances, showing how initial conditions interact with emergent events to shape relationship trajectories toward either strengthening or dissolution. Our study applies and extends this developmental perspective by mapping failure onto specific lifecycle stages in the Helix context and examining how the transition from interpersonal relationships to institutionalised arrangements (a transition many collaborations never complete) constitutes a critical vulnerability.

The organisational failure literature offers further theoretical resources. Mellahi and Wilkinson (2004) distinguish between deterministic perspectives that emphasise environmental factors beyond organisational control and voluntaristic perspectives that focus on managerial choices and organisational weaknesses. They argue for an integrative approach that examines how external pressures interact with internal vulnerabilities, an approach we adopt. The institutional theory tradition similarly highlights how organisations embedded in different institutional logics may struggle to collaborate effectively when these logics conflict (Thornton, Ocasio & Lounsbury, 2012). Universities operate under academic logics prioritising research excellence and scholarly publication; firms operate under commercial logics emphasising profit and market share; government agencies operate under public accountability logics focused on policy outcomes. When these logics conflict, as they frequently do, collaboration may become fraught or unsustainable. Importantly, institutional logics are not merely different preferences: they shape temporal orientations, definitions of success, and accountability structures in ways that are difficult to bridge through goodwill alone.

2.3 *Factors Contributing to Collaboration Breakdown*

Drawing on the literature reviewed above, we identify four categories of factors that may contribute to the failure of collaboration. These categories are not mutually exclusive, and our empirical analysis will trace how such interactions produce cascading failure processes.

Relational factors concern the quality of relationships among partners. Trust is widely recognised as foundational to successful collaboration (Vangen & Huxham, 2003). When trust breaks down, whether through perceived betrayal, unmet expectations, or accumulated grievances, collaboration becomes difficult to sustain. Communication failures, interpersonal conflicts, and power

imbalances that leave some partners feeling marginalised or exploited also fall within this category.

Structural factors concern the governance arrangements and resource base of the collaboration. Unclear roles and responsibilities, inadequate governance mechanisms, insufficient resources, and over-reliance on key individuals rather than on institutionalised processes can all undermine the sustainability of collaboration. The literature on boundary spanners highlights how individual champions often hold collaborations together through personal relationships and cross-institutional knowledge (Williams, 2002); when such individuals depart, collaborations that have not developed institutionalised alternatives are acutely vulnerable.

Institutional factors concern the compatibility of partners' institutional contexts. Conflicting institutional logics- for example, academic versus commercial priorities- create ongoing tensions that manifest in different definitions of success, different time horizons for results, and different accountability structures. Misaligned incentive structures, where collaboration effort is not recognised or rewarded within partners' home institutions, discourage sustained engagement (Thornton et al., 2012). These misalignments are often latent at project initiation, only becoming visible as the collaboration encounters real operational pressures.

Environmental factors concern external developments beyond the collaboration's direct control (Bryson et al., 2015). Critically, environmental factors rarely operate alone: they expose and amplify latent structural and relational vulnerabilities, transforming manageable weaknesses into crisis conditions. Funding transitions, leadership changes in partner organisations, and external shocks such as the COVID-19 pandemic are particularly consequential in this regard, as each can rapidly destabilise collaborations that lack robust governance or diversified resource bases.

2.4 *Conceptual Framework*

Building on this review, we propose a conceptual framework for analysing collaboration failure that integrates two dimensions: the type of failure and the stage at which failure occurs. We design this framework as an analytical structure; consequently, our empirical analysis allows us to assess which failure types, and which combinations of factors characterise the cases we study.

Regarding types of failure, we distinguish five forms: First, *Dissolution* refers to collaborations that formally ended, with structures dismantled and activities ceased. Second, *Zombie collaboration* refers to partnerships that formally exist but exhibit no meaningful activity on paper and are effectively defunct. Third, *Stillborn collaborations* are those that were initiated but never achieved

operational status. Fourth, *Fragmentation* occurs when core partners withdraw, leaving a significantly diminished collaboration that may continue in reduced form, and Fifth, *Mission drift* describes collaborations that continue but have abandoned their original purpose, potentially indicating failure to achieve initial objectives.

These types are analytically distinct, but their boundaries warrant brief clarification. Dissolution and fragmentation both involve endings, but dissolution implies complete cessation whereas fragmentation implies partial continuation under a reduced partner configuration. Zombie collaboration and mission drift both describe collaborations that persist without genuine vitality. Still, zombie collaboration is characterised by inactivity and the absence of meaningful engagement, while mission drift is characterised by misdirected activity and continued operation toward substituted rather than original objectives. Stillborn collaborations are distinguished from early dissolution by the absence of an operational phase: they fail before meaningful joint activity begins rather than collapsing after it has commenced. These distinctions matter analytically because different failure types implicate distinct intervention points, contributing factors, and forms of organisational learning.

Regarding stages, we draw on Ring and Van de Ven's (1994) developmental model of inter-organisational relationships, already introduced in Section 2.2, to distinguish three analytically separable but empirically overlapping phases: *formation* (initiation, negotiation, agreement), *operation* (activity delivery, governance), and *institutionalisation* (embedding, routinisation, succession). This three-phase structure is consistent with broader process perspectives on collaborative development (Huxham & Vangen, 2005). It provides a temporal scaffold for locating when different failure types are most likely to emerge. Collaborations may stall, regress, or terminate at any point across these phases, but certain types of failure are more characteristic of stages.

Stillborn collaborations fail at formation, when foundational incompatibility or resource shortfalls prevent the partnership from becoming operational. Dissolution and fragmentation typically occur during operation, when governance is actively tested and structural or relational weaknesses become visible under the pressures of real joint activity. Zombie collaboration is primarily a relational failure of the operation stage: partners continue to meet and report formally. At the same time, meaningful engagement and shared purpose have already eroded; the collaboration persists in appearance but not in substance. This distinction from dissolution is important – zombie collaborations have not ended; they have hollowed out from within. Mission drift, by contrast, is characteristically a failure at the institutionalisation stage. It emerges when a collaboration survives long enough to seek embedding and succession but

TABLE 1 Conceptual Framework for Analysing University–Stakeholder Collaboration Failure

Contributing Factors		Stage 1: Formation Initiation · Negotiation · Agreement	Stage 2: Operation Activity Delivery · Governance	Stage 3: Institutionalisation Embedding · Routinisation · Succession	Cascading Failure Process
RELATIONAL	Trust erosion Communication breakdown Power imbalance Partner marginalisation Declining engagement	**High salience** Trust-building critical at formation stage → *Stillbirth risk*	**High salience** Gradual erosion; power asymmetries emerge → *Fragmentation / Zombie*	**Moderate** Accumulated damage impedes embedding → *Mission Drift / Dissolution*	*Latent vulnerabilities exposed when environmental triggers hit*
STRUCTURAL	Weak governance Key-person dependency Resource fragility No succession planning Unclear roles	**Moderate** Governance design deferred; latent weakness created → *Stillbirth* (*if extreme*)	**High salience** Governance tested; key-person exits → *Fragmentation / Dissolution*	**High salience** Failure to institutionalise beyond founding team → *Zombie / Mission Drift*	↓
INSTITUTIONAL	Conflicting logics Incentive misalignment Temporal mismatch Ambiguous success criteria Accountability gaps	**Moderate** Logic conflicts rarely negotiated at outset → *Stillbirth* (*if irreconcilable*)	**High salience** Incentive conflicts surface under operational pressure → *Fragmentation / Mission Drift*	**Moderate** Reward systems fail to sustain engagement → *Zombie / Dissolution*	↓
ENVIRONMENTAL	Funding transitions Leadership turnover External shocks (e.g. COVID) Policy changes Economic pressures	**Low** Context shapes initiation conditions	**Moderate** Shocks expose structural and relational vulnerabilities	**High salience** Funding ends; triggers cascading breakdown → *Zombie / Dissolution*	↓
FAILURE TYPES		**Stillbirth** *Fails at formation stage*	**Fragmentation · Dissolution** *Fails during operation*	**Zombie · Mission Drift** *Fails at institutionalisation*	**Collapse**

Note. Salience levels (High / Moderate / Low) indicate the relative importance of each factor at each lifecycle stage. Factors interact; combinations produce cascading effects. Environmental factors operate primarily as proximate triggers exposing latent structural and relational vulnerabilities.

does so by abandoning its founding objectives to secure continued funding or political support, thereby substituting organisational survival for its original purpose. These distinctions are analytically important because different failure types implicate distinct intervention points and combinations of factors.

Our framework positions the four factor categories: *relational, structural, institutional*, and *environmental* as potential contributors to failure at each stage, with the expectation that different factors will be more salient at different stages. Relational factors may be particularly important during formation, when trust is being established; structural factors become more salient during operation, when governance is tested; and environmental factors may trigger crises during attempts at institutionalisation, when external sustainability is being sought. However, we treat these as sensitising expectations rather than fixed hypotheses. The framework is presented schematically in *Table 1 below* and applied systematically in the cross-case analysis presented in *Table 5*.

The framework also anticipates that different failure types are associated with characteristic process trajectories. Stillborn collaborations fail rapidly at formation due to incompatibility; dissolution and fragmentation follow extended deterioration processes during the operational phase; zombie collaboration represents a prolonged process of hollowing out, in which surface activity persists while internal commitment erodes; and mission drift reflects a slow substitution of the original purpose during institutionalisation attempts. Understanding failure type therefore also implies understanding the temporal and processual character of the breakdown, which Section 4.5 examines in detail.

3 Methodology

The study is grounded in interpretivist and constructivist epistemology, acknowledging that failure is not merely a technical outcome but a socially constructed, retrospective, and often contested process (Guba & Lincoln, 1994; Dana & Dumez, 2015). Participants interpret and narrate failure through their own institutional lenses, and these interpretations are themselves data. Following abductive logic (Timmermans & Tavory, 2012), we iteratively moved between empirical material and conceptual framing to develop grounded yet theoretically informed insights. Our analytical approach draws primarily on the Gioia methodology (Gioia et al., 2013) to structure the data hierarchy (Table 4), while employing the process-oriented, cross-case comparison characteristic of Eisenhardt's (1989) case study approach to identify patterns across cases. Gehman et al. (2018) demonstrate that these approaches are compatible when

the researcher's goal is theory elaboration rather than pure theory generation, which is precisely our aim here. Critically, our prior conceptual framework of four contributing factor categories served as a sensitising device (Blumer, 1954) rather than a deductive template: it oriented our attention without predetermining what we would find, leaving the coding process open to empirical categories not anticipated by the framework. Consistent with this epistemological positioning, causal claims in this study are understood as processual and explanatory rather than deterministic: we identify mechanisms and interaction patterns that appear to generate failure trajectories, not universal laws that predict failure outcomes.

3.1 *Case Selection*

We employed purposive sampling (Patton, 2015) to identify three collaborative projects that varied in failure type, partner configuration, and institutional structure. The unit of analysis in this study was the failed university–stakeholder collaboration rather than the individual organisation or participant. Potential cases were first identified through outreach to university innovation offices, regional development agencies, business support organisations, and intermediary actors in Finland. This process identified approximately 12 collaborations that had either formally ended, become inactive, fragmented, or substantially drifted from their original purpose over the preceding 5 years.

From this initial pool, cases were screened against four inclusion criteria: (a) involvement of at least one university and one external stakeholder; (b) operational duration of at least two years, to exclude very early-stage failures where insufficient organisational learning has occurred; (c) evidence of failure in any form consistent with our operational definition (see Section 2.2); and (d) availability of at least two participants per case with direct involvement and retrospective insight. Three cases met all criteria and were selected because together they provided variation in failure trajectories, stakeholder compositions, and Helix configurations while sharing the common outcome of collaboration breakdown.

As shown in Table 2, Project A was a regional innovation hub that transitioned from zombie status to eventual dissolution; Project B was an academic–industry R&D partnership that fragmented as key partners withdrew; and Project C was a joint civil society–university innovation intermediary that experienced mission drift before dissolving. Projects A and B primarily reflected Triple Helix arrangements. In contrast, Project C included a civil society partner and therefore enabled analysis of Quadruple Helix dynamics, specifically whether formal civil society participation generated collaboration dynamics that differed from the Triple Helix-oriented arrangements observed in the other two cases.

The study prioritised analytical depth and process reconstruction over empirical breadth. Given the exploratory aim of examining collaboration failure processes, an area that remains under-researched in the Helix literature, we adopted a theoretically oriented case study logic rather than seeking statistical representativeness (Eisenhardt, 1989; Yin, 2018). Although the empirical base is modest, each case generated rich retrospective accounts, complemented by documentary evidence, enabling triangulation and the chronological reconstruction of the breakdown of collaboration. The goal of the study is therefore theory elaboration and conceptual refinement rather than universal causal generalisation.

In Yin's (2018) terms, these cases function as revelatory cases: they provide access to a phenomenon, namely Helix collaboration failure, that has rarely been examined systematically and that existing theory is not well equipped to explain. Their theoretical value lies not in their representativeness but in their capacity to generate conceptual insights that can be developed and tested in future research across broader empirical contexts.

TABLE 2 Overview of Cases

Case	Type & Focus	Duration	Failure Type	Lead Partner	Main Funding Structure	Partner Composition
Project A	Regional innovation hub	2016–2021	Zombie → Dissolution	University A	EU regional development funding	University A; Regional Development Agency; 3 SMEs; Chamber of Commerce
Project B	Academic–industry R&D partnership	2018–2022	Fragmentation → Zombie	University B + Industry Federation	Public innovation and partner co-funding	University B; 2 Technology Firms; Industry Federation; Municipal Government
Project C	Civil society–university intermediary	2019–2023	Mission Drift → Dissolution	Intermediary Body / University C	Regional government project funding	University C; Civil Society Organisation; Regional Government Agency; Intermediary Body

SOURCE: DEVELOPED BY THE AUTHORS

3.2 *Data Collection*

Data collection relied on two primary sources: semi-structured interviews and documentary evidence. We conducted six semi-structured interviews between October 2025 and February 2026, with two interviews per case. Each interview lasted 40–60 minutes and was held online. Interviews were audio-recorded with participant consent and transcribed verbatim. Participants were purposively selected to represent different partner types (university, industry, government, civil society) and different roles within each collaboration. Table 3 provides an overview of participants.

TABLE 3 Overview of Interview Participants

Code	Case	Participant Role	Organisation Type	Role in Collaboration
P1	**Project A**	Innovation Manager	University (HEI)	Coordinated university participation and regional engagement activities
P2	**Project A**	Liaison Officer	Business Support Organisation	Facilitated communication between SMEs, regional actors, and university partners
P3	**Project B**	Director of Engagement	University (HEI)	Oversaw institutional collaboration strategy and partnership coordination
P4	**Project B**	Programme Manager	Regional Development Agency	Managed regional development objectives, funding expectations, and stakeholder reporting
P5	**Project C**	Research Coordinator	University (HEI)	Coordinated research and co-design activities with external partners
P6	**Project C**	Project Lead	Civil Society Intermediary	Represented civil society interests and coordinated community-facing activities

Note. Participants were selected to represent different partner types and roles within each collaboration. Participant Role indicates the formal position held within the collaboration; Organisation Type indicates the institutional sector the participant represented; Role in Collaboration describes the primary function each participant performed within the partnership. All names and organisational identities have been anonymised. Interview duration ranged from 40 to 60 minutes.

SOURCE: DEVELOPED BY THE AUTHORS

We conducted six semi-structured interviews, two per case, each lasting 40–60 minutes. While modest in number, this sample reflects the deliberate prioritisation of depth over breadth appropriate to retrospective failure research, where access is constrained, and rich narrative accounts are more analytically valuable than larger but thinner datasets. Data collection continued until thematic saturation was reached, with no substantively new failure mechanisms emerging across the final interviews (Guest et al., 2006). Interview data were complemented by documentary evidence for each case, including founding agreements, progress reports, and strategic plans, which supported triangulation and timeline reconstruction (Denzin, 1978; Dana & Dumez, 2015).

Interview questions addressed five themes: *project formation and initial expectations; operational experiences; turning points and warning signs; explanations for failure; and aftermath and counterfactuals*. Questions were open-ended to allow participants to construct their own narratives, with probing follow-ups on specific events and decisions. All participants provided informed consent, were assured of confidentiality, and organisational identities were anonymised (Kaiser, 2009).

3.3 *Data Analysis*

We employed an abductive, process-oriented multiple case analysis to identify the mechanisms and interacting variables underlying collaboration failure. Consistent with abductive reasoning (Timmermans & Tavory, 2012), the analysis iteratively moved between empirical material and existing theoretical perspectives on inter-organisational collaboration, institutional logics, and organisational failure. Rather than treating theory as a fixed explanatory template, the analytical process involved continuous dialogue between emergent empirical observations and sensitising theoretical concepts, enabling progressive refinement of interpretation throughout the study.

The analytical process proceeded in five interconnected stages.

We began with in-depth within-case analyses to construct detailed process narratives for each collaboration project. Following established case study procedures (Eisenhardt, 1989; Eisenhardt & Graebner, 2007; Ring & Van de Ven, 1994), we chronologically reconstructed each project's lifecycle, identifying key phases, tensions, and breakdown points. Particular attention was given to critical incidents such as funding transitions, leadership exits, governance restructuring, and shifting strategic priorities, which were examined as potential turning points that exposed underlying structural and relational vulnerabilities.

Second, we conducted open coding inductively using interview transcripts, documentary evidence, analytic memos, and spreadsheet-based coding matrices to ensure transparency and traceability between raw data and conceptual

abstraction. Initial coding remained intentionally close to participants' language to preserve informant-centric meaning (Gioia et al., 2013). Examples of first-order concepts included expressions such as "nobody took ownership," "we were running on goodwill," "the timelines never matched," and "meetings became transactional." At this stage, coding was conducted without imposing the prior conceptual framework, allowing themes to emerge directly from the empirical material.

Third, through iterative comparison within and across cases, first-order concepts were progressively consolidated into second-order dimensions reflecting recurring organisational patterns and mechanisms. Only at this stage were the four sensitising factor categories introduced: relational, structural, institutional, and environmental. These categories functioned heuristically rather than deductively; that is, they oriented interpretation without constraining it. We assessed whether emerging dimensions mapped onto, extended, or challenged the prior framework.

Importantly, not all emerging themes aligned neatly with the initial framework. For example, the aggregated theme of Cascading Vulnerability and Collapse emerged inductively from the data and was not anticipated by the original framework. This demonstrated that the analytical process remained open to theoretically unexpected patterns rather than merely confirming prior assumptions. The resulting data structure is presented in Table 4.

TABLE 4 Data Structure

First-Order Concepts	Second-Order Dimensions	Aggregated Themes
Meetings becoming transactional; attendance driven by obligation; slower communication responses; questioning relevance of collaboration; decline in openness and information sharing; feeling reduced to implementation role; gradual erosion of shared purpose	Declining collaborative engagement; erosion of trust and reciprocity	**Relational Deterioration**
Reliance on informal coordination; absence of clear governance framework; unclear responsibility allocation; circulating rather than resolving problems; dependence on bridging individual; leadership departure weakening collaboration; lack of long-term funding plan; assumed continuation of external support	Weak governance structures; key-person dependency; resource dependence and fragility	**Structural Vulnerability**

TABLE 4 Data Structure (*cont.*)

First-Order Concepts	Second-Order Dimensions	Aggregated Themes
End of external funding period; pandemic-related disruption; leadership turnover; policy and strategic shifts; vulnerabilities exposed during shocks	Funding transition; external shocks; strategic destabilisation	**Environmental Disruption**
Warning signs interpreted as temporary; increasing effort required to sustain activity; gradual loss of momentum; collaboration becoming inactive without formal closure	Escalating disengagement; informal dissolution dynamics	**Cascading Vulnerability and Collapse**
Publication-based evaluation systems; difficulty prioritising collaboration activities; conflicting institutional priorities; expectation of short-term measurable outputs; long-term academic project orientation; short funding cycles; undefined performance expectations	Incentive misalignment; temporal misalignment; ambiguous success criteria	**Institutional Misalignment**

SOURCE: DEVELOPED BY THE AUTHORS

Fourth, following within-case interpretation, we conducted systematic cross-case comparison using cross-case matrices, temporal mapping, and pattern-matching techniques (Eisenhardt, 1989). We compared how similar mechanisms manifested differently across institutional contexts, governance arrangements, stakeholder compositions, and collaboration stages. Attention was given to identifying recurring deterioration sequences, interaction effects among factor categories, and the role of critical junctures in accelerating decline.

The cross-case analysis focused on two analytical questions. First, which contributing conditions appeared consistently across all three cases and which appeared contingent on specific institutional contexts? Second, how did combinations of relational, structural, institutional, and environmental factors interact dynamically over time to produce cascading failure processes rather than isolated breakdown events? This process-oriented comparison enabled us to move beyond static explanation toward analysis of failure trajectories and interaction mechanisms.

Finally, we returned iteratively to the relevant theoretical literature to refine conceptual interpretation and theoretical contribution. This iterative

movement between data and theory is consistent with abductive analysis, where empirical anomalies and recurring patterns progressively reshape conceptual understanding (Timmermans & Tavory, 2012). To enhance analytical credibility, we triangulated interview accounts with documentary evidence, maintained reflexive analytic memos throughout interpretation, and engaged in ongoing analytic discussion to critically assess emerging interpretations and surface potential researcher assumptions or biases (Charmaz, 2014).

4 Findings

Through the three collaboration projects, failure unfolded gradually rather than abruptly. Participants consistently described breakdown as a process marked by relational fatigue, structural vulnerability, institutional misalignment, and environmental disruption, which together constitute the four categories of our conceptual framework. Importantly, no case was characterised by a single dominant cause; in every instance, failure emerged from the interaction of multiple factors. What follows presents the four dimensions thematically, drawing on evidence across cases, before Section 4.5 examines the dynamic process through which these factors interacted to produce collapse. Table 5, presented at the end of this section, provides a systematic cross-case comparison that maps each case onto our conceptual framework.

4.1 *Relational Erosion*

Relational deterioration was present in all three cases and constituted the most visible dimension of decline for participants. However, it is rarely manifested as dramatic conflict. Instead, participants described a gradual erosion of relational energy, with engagement diminishing incrementally over time. In Projects A and C, civil society and public sector partners felt progressively marginalised. This erosion was particularly pronounced. A senior academic in Project A reflected:

> In the first year, there was a real sense that we were building something together. People were genuinely curious about each other's perspectives. But after a while, that curiosity faded. Meetings became more transactional; we were reporting updates rather than thinking collectively. You could sense that people were attending because they felt they had to, not because they were invested. (P1)

This account illustrates how relational dynamics shifted from collaborative enthusiasm to procedural obligation, a shift that, once initiated, proved

difficult to reverse. In Project A, a specific incident accelerated this process. A university participant described a moment when trust began to erode:

> There was one situation where a smaller group prepared a funding application without fully informing the rest of us. It wasn't necessarily malicious, but it created uncertainty. After that, people were more careful about what they shared. The openness we had in the beginning wasn't quite the same. (P1)

In Project C, the Quadruple Helix case involving civil society, partner marginalisation was more structurally embedded from the outset. The civil society intermediary partner described an experience of systematic exclusion from meaningful decision-making:

> At some point, we started feeling like implementation partners rather than equal collaborators. Decisions seemed to be shaped before we were consulted. It wasn't confrontational, but you could feel that our influence had decreased. (P6)

This finding is significant for Quadruple Helix theory. If genuine QH collaboration requires substantive civil society inclusion (Carayannis & Campbell, 2009), then Project C's failure may partly reflect the persistence of Triple Helix power dynamics even within formally Quadruple Helix arrangements. The civil society partner's experience of marginalisation, being treated as an implementation agent rather than a co-creator, undermined not only their individual commitment but the collaborative legitimacy of the project. Across cases, relational breakdown was cumulative: no single rupturing caused collapse, but small shifts accumulated until trust and shared purpose were sufficiently eroded.

Across cases, relational erosion followed a consistent sequence: early enthusiasm gave way to procedural obligation, then to selective disengagement, and finally to a legitimacy deficit. What differed was the primary driver. In Project A, this was a betrayal of trust through exclusion from a funding decision; in Project B, it was a breakdown in coordination following a key departure; in Project C, it was the structural marginalisation of the civil society partner. The relational trajectory was therefore similar across cases while its origins were case-specific.

4.2 *Structural Fragility*

Structural weaknesses were present across all three cases. Still, they were most consequential in Projects A and B, where governance informality and

key-person dependency left collaborations unable to adapt when personnel or resources changed. A university project manager in Project A explained:

> We relied heavily on informal coordination. If something needed to move forward, we would call the person we knew best. That worked while relationships were strong. But when roles changed, we realised there wasn't a clear governance framework holding everything together. (P1)

Another participant described the ambiguity surrounding responsibility that this informality produced:

> When delays started happening, it wasn't clear who was accountable. Was it the university? The regional agency? The project board? Because responsibilities weren't very formalised, problems tended to circulate rather than be resolved. (P2)

Key-person dependency was the critical structural vulnerability in Project B. An external stakeholder spoke directly about its consequences:

> One particular colleague acted as a bridge between all of us. She understood the academic environment and the policy side. When she left, the connection weakened immediately. We tried to redistribute tasks, but it wasn't just about tasks, it was about relationships and continuity. (P4)

This reflects a broader pattern identified in the boundary-spanning literature (Williams, 2002): when a single individual embodies the social capital that holds a multi-institutional collaboration together, their departure is not merely a staffing problem but a structural collapse. Project B's failure accelerated markedly following this exit, suggesting that the collaboration had never successfully institutionalised beyond the founding relationships. Funding structure also exposed fragility across cases as Project A, Innovation Manager points out:

> We had assumed that continuation funding would follow because the project was promising. But there was no concrete sustainability plan. When the initial funding ended, we suddenly realised how dependent we had been on that external support. (P1)

Structural fragility, therefore, did not prevent initial success; early momentum was achievable through enthusiasm and informal coordination. But it fundamentally undermined resilience when pressures intensified.

Across cases, structural fragility operated as a background vulnerability rather than an immediate cause of failure. Projects A and B exhibited the most acute structural weaknesses, with governance informality and key-person dependency leaving both collaborations unable to absorb changes in personnel and resources. Project C's structural fragility was more moderate but interacted with environmental disruption to produce the same cascading outcome, suggesting that even partial governance development offers limited protection when environmental shocks are severe.

4.3 *Institutional Misalignment*

Institutional differences surfaced most clearly around incentives and timelines and were present to some degree in all three cases, though most acutely in Project B. A senior academic explained the contradiction at the heart of university participation:

> Within the university, collaboration is encouraged rhetorically, but performance evaluation is still largely publication based. When workloads increased, it became difficult to justify spending significant time on partnership activities that were not directly recognised. (P3)

An industry representative in Project B described divergent expectations that were never openly negotiated:

> From our side, we were expecting tangible outputs within a year of pilot tools or measurable improvements. But the academic team was thinking in terms of research design, ethics approvals, and multi-year trajectories. We weren't frustrated with them as individuals, but the systems we operate in are very different. (P4)

A business support representative noted the quiet pressure created by temporal mismatch:

> Our organisation works on short funding cycles. We need visible outcomes quickly to justify continuation. The university partners were more comfortable with longer horizons. That difference created quiet pressure. (P2)

In Project C, institutional misalignment took a different form. The civil society intermediary operated under community accountability logic, requiring responsiveness to local needs and participatory processes, while the university prioritised academically legitimate outputs and the regional government

agency sought policy-relevant deliverables. None of these logics was inherently incompatible, but the absence of explicit negotiation meant each partner gradually pursued their institutional priorities at the expense of shared objectives. A university participant reflected:

> In hindsight, we never had a detailed conversation about what 'success' meant for each partner. We assumed alignment because we shared a general vision, but our institutional incentives were pulling us in different directions. (P5)

These institutional mismatches were rarely openly debated. Instead, they manifested gradually through declining enthusiasm, shifting priorities, and ultimately disengagement, a pattern consistent with Thornton et al.'s (2012) account of how institutional logic conflicts produce organisational strain.

The cross-case pattern suggests that institutional misalignment operated differently depending on which logic conflicted. In Project B, the academic-commercial divide was the dominant fault line. In Project C, three distinct logics, namely academic, community accountability, and policy delivery, are pulled simultaneously in different directions, producing a more diffuse but equally destabilising form of misalignment. In Project A, institutional tensions were present but subordinate to structural and environmental factors, indicating that misalignment's destructive potential depends in part on which other vulnerabilities are active simultaneously.

4.4 *Environmental Disruption*

Environmental factors functioned differently across cases. In Project A, the end of the EU funding period was the proximate trigger; in Project B, the departure of a key industry partner coincided with organisational restructuring; in Project C, the COVID-19 pandemic disrupted an already fragile collaboration. What was consistent across cases was that environmental factors did not operate independently. They exposed and amplified pre-existing vulnerabilities. A regional development actor in Project A described the funding transition:

> The end of the EU funding period was a critical moment. Until then, the collaboration had a clear structure and budget. When that security disappeared, the conversation shifted from innovation to survival. That changed the dynamic immediately. -(P2)

A university leader discussed leadership turnover as an environmental trigger:

> When senior management changed, regional engagement was no longer emphasised in the same way. The collaboration wasn't terminated, but it lost strategic visibility. Without strong internal backing, momentum slowed. -(P_3)

In Project C, the pandemic played a catalytic role:

> COVID forced everyone to reprioritise. Universities focused on teaching continuity; businesses focused on survival. The collaboration became secondary. Once the rhythm was broken, it was difficult to re-establish. -(P_5)

Critically, another stakeholder in Project C articulated the interaction between structural vulnerability and environmental shock. The core mechanism of cascading failure:

> If the collaboration had been more institutionalised, it might have survived those changes. But because it depended on specific people and funding streams, external shifts had a disproportionate impact. -(P6)

The cross-case comparison reveals an important distinction in how environmental factors operated. In Project A, the funding transition was predictable and foreseeable, yet no sustainability plan had been developed, pointing to a governance failure rather than an unforeseeable shock. In Project B, the environmental trigger was partly endogenous, as the industry partner's restructuring reflected tensions that had been built within the collaboration. In Project C, the COVID-19 pandemic was genuinely external and unpredictable, yet its impact was magnified by pre-existing relational and structural fragility. This variation suggests that environmental disruption is best understood not as a uniform external force but as a catalyst whose destructive potential depends on the depth of vulnerability already present within the collaboration.

4.5 *The Dynamics of Failure: Critical Junctures, Cascading Effects, and Collapse*

Across all three cases, failure was described not as an event but as a process of gradual accumulation of unresolved tensions, punctuated by critical junctures that accelerated the decline. This dynamic is the most theoretically significant finding of our study and merits extended analysis.

Each collaboration followed a broadly similar trajectory. An initial period of relative success and enthusiasm, characterised by active participation, open communication, and a genuine collaborative spirit, gave way to a middle phase of gradually declining engagement, which participants at the time interpreted as temporary fluctuation. Warning signs were present but systematically misread. An academic in Project A reflected:

> Looking back, there were warning signs for lower attendance, less preparation, fewer new ideas. But at the time, we interpreted those as temporary fluctuations. It was only later that we realised they were part of a broader pattern. -(P1)

This misreading of warning signs was not simply inattention. It reflects a systematic cognitive tendency that collaboration literature might call 'collaborative optimism bias': participants invested in a partnership have strong incentives to interpret ambiguous signals positively and avoid the reputational and relational costs of raising concerns. By the time the pattern became undeniable, recovery had typically become structurally impossible.

Critical junctures, discrete events that crystallised latent tensions, were identifiable in all three cases. In Project A, the end of EU funding was the critical juncture, revealing that no sustainability model had been developed. In Project B, the departure of the key boundary spanner was the crystallising event. In Project C, the pandemic disruption exposed both the absence of a resilient governance structure and the accumulated relational damage of partner marginalisation. What each critical juncture shared was that it did not create the collaboration's problems – it revealed them.

The causal mechanism at work across all three cases can therefore be described as follows. Latent structural and relational vulnerabilities accumulated gradually during the operation phase, remaining largely invisible because collaborative optimism bias led participants to interpret warning signs as temporary. When a critical juncture occurred, whether a funding transition, a key departure, or an external shock, it removed the conditions that had sustained surface-level functioning and exposed the accumulated fragility beneath. The speed of subsequent cascading breakdown was directly proportional to the depth of latent vulnerability: the more governance had been deferred, and relationships had eroded, the more rapidly the critical juncture translated into irreversible decline.

Following critical junctures, cascading effects were rapid. Initial problems damaged trust, which impeded communication, which prevented problem-solving, which deepened distrust and further eroded commitment. A stakeholder described the emotional dimension of this final phase:

> There comes a moment when you realise that keeping the collaboration alive requires more energy than anyone is willing to invest. It's not dramatic; it's just exhaustion. -(P2)

Notably, none of the three collaborations experienced formal closure. They dissolved through progressive inactivity, effectively becoming zombie collaborations before any official termination. A participant observed:

> No one formally declared the collaboration over. It just became inactive. After several months without meaningful activity, it was clear that it had effectively ended. -(P1)

This absence of formal closure has practical implications. Without explicit reflection and organisational learning from failure, participants carry unresolved narratives and unreflected-upon lessons into subsequent attempts at collaboration. A university representative concluded:

> I wouldn't call it a failure in a dramatic sense. It simply didn't sustain itself. The initial conditions that made it possible were no longer there. -(P1)

This framing of failure as the absence of sustaining conditions, rather than as active breakdown, is both an empirical observation and an analytical insight. It suggests that collaborative resilience is not built by avoiding failure but by actively constructing the institutional, structural, and relational conditions that enable collaborations to persist amid inevitable turbulence.

Table 5 presents a systematic cross-case comparison that maps our empirical findings onto the conceptual framework. Several patterns are notable. Structural vulnerability was uniformly high across all three cases, suggesting it serves as a prominent, recurring background condition for failure in this context. Environmental triggers were also present in all three cases, but their severity varied; the collaboration that experienced the most severe environmental shock (Project C, COVID-19 plus leadership change) was also the one with the least robust governance. Institutional misalignment was most acute in Project B, where it functioned as the primary driver of fragmentation rather than merely a contributing factor. Relational erosion was present in all cases but was most consequential in Projects A and C, where partner marginalisation created legitimacy deficits that weakened the collaborative foundation. Finally, the absence of civil society voice in Projects A and B – compared to its marginalised presence in Project C – suggests that the mere inclusion of additional helix actors does not guarantee the equitable participation that Quadruple Helix theory advocates.

TABLE 5 Cross-Case Comparison: Applying the Conceptual Framework

Dimension	Project A	Project B	Project C
Failure Type	Zombie → Dissolution	Fragmentation → Zombie	Mission Drift → Dissolution
Failure Stage	Operation → attempted institutionalisation	Operation	Institutionalisation attempt
Primary Trigger	EU funding end + leadership departure	Key individual exit + industry retrenchment	Leadership turnover + pandemic disruption
Relational Factors	High (trust erosion central)	Moderate (coordination breakdown)	High (partner marginalisation)
Structural Factors	High (governance vacuum, key-person dependency)	High (no succession planning)	Moderate (informal coordination)
Institutional Factors	Moderate (timeline mismatch)	High (incentive misalignment dominant)	High (mission creep under policy pressure)
Environmental Factors	High (funding cycle as trigger)	Moderate (economic pressure on industry partner)	High (COVID + leadership change combined)
Recovery Attempted?	Yes – failed due to no sustainability plan	Partial – reduced collaboration continued briefly	Yes – rebranding attempted; objectives abandoned
Civil Society Voice	Absent	Absent	Present but marginalised

SOURCE: DEVELOPED BY THE AUTHORS

5 Discussion

This study set out to examine why university–stakeholder collaborations fail and what conditions and mechanisms lead to their dissolution. Four findings stand out, each with distinct theoretical implications elaborated below. Briefly: failure was consistently multicausal and processual; structural vulnerability

operated as a persistent background condition while environmental triggers served as proximate catalysts; failure followed a recognisable trajectory from latent decline through critical juncture to informal dissolution; and formal civil society inclusion in Quadruple Helix arrangements did not produce the equitable participation that distinguishes genuine QH collaboration.

5.1 *Theoretical Contributions*

This study makes several contributions to the Helix and inter-organisational collaboration literature. First, by systematically examining failed collaborations and applying our failure typology to empirical cases, we address the survivorship bias that characterises much existing Helix research, which has predominantly focused on functioning or successful collaborations. Our findings suggest that the mere presence of university, industry, and government actors is insufficient to sustain collaboration over time. Instead, collaborative durability appears to depend on the interaction of governance adequacy, institutional alignment, relational resilience, and the capacity to absorb environmental disruption.

More broadly, our findings suggest that Helix collaborations may be better understood not as stable institutional arrangements but as dynamically fragile organisational processes whose sustainability depends on successful transition across lifecycle stages. Existing Helix research has primarily focused on actor configuration and collaboration formation, while paying comparatively limited attention to how collaborations become institutionalised over time. Our cases suggest that the transition from relationship-based coordination to institutionalised governance constitutes a critical vulnerability point. Collaborations may initially function through enthusiasm, interpersonal trust, and informal coordination, yet remain structurally fragile if governance routines, succession mechanisms, and sustainability arrangements fail to consolidate. This shifts analytical attention from collaboration formation toward collaboration durability and resilience, thereby extending Helix theory in a more process-oriented direction.

The concept of structural vulnerability as used here draws on and extends governance research in inter-organisational settings. Provan and Kenis (2008) distinguish between participant-governed and lead organisation-governed networks, showing that governance form shapes network capacity to absorb stress; our findings suggest that in Helix collaborations, the failure to transition from informal interpersonal coordination to any formalised governance mode constitutes the central structural vulnerability, regardless of which governance form is adopted.

Second, existing Helix research has treated collaboration outcomes as essentially binary, success or failure, obscuring the analytically important variation in how failure unfolds. Our typology of failure (dissolution, zombie collaboration, stillbirth, fragmentation, and mission drift) addresses this limitation directly. The empirical application of this typology in Table 5 suggests that different forms of breakdown are associated with different configurations of contributing conditions: fragmentation in Project B appeared closely associated with institutional misalignment and structural dependency, whereas mission drift in Project C reflected the interaction of all four factor categories under conditions of prolonged environmental disruption. Rather than treating collaboration failure as a single outcome, our findings suggest that it unfolds along multiple trajectories that require more differentiated analytical treatment. A distinction that has practical as well as theoretical implications for how researchers and practitioners diagnose and respond to collaborative decline.

Third, Helix theory has been criticised for its relatively static treatment of collaboration dynamics, focusing on actor configuration rather than how partnerships evolve, deteriorate, and potentially collapse over time (Cai & Lattu, 2022). Our findings extend this literature by showing that failure is not a discrete endpoint but a cumulative process: collaborations moved through recognisable periods of enthusiasm, gradual deterioration, and eventual destabilisation, punctuated by critical junctures that accelerated decline rather than causing it. Unresolved tensions progressively reduced collaborative resilience long before any formal breakdown occurred.

Relatedly, our findings refine inter-organisational collaboration and organisational failure theory by demonstrating how environmental disruptions interact with latent vulnerabilities rather than independently causing collapse. Across various cases, funding transitions, leadership turnover, and the COVID-19 pandemic did not alone produce failure. Instead, these disruptions appeared to expose accumulated weaknesses in governance, institutional alignment, and relational cohesion that had gradually developed during earlier stages of collaboration. This interactional dynamic helps bridge deterministic and voluntaristic perspectives on organisational failure (Mellahi & Wilkinson, 2004), suggesting that the breakdown of collaboration emerges from the interaction between external shocks and unresolved internal tensions.

Fourth, our findings contribute to debates regarding the added value and governance implications of Quadruple Helix arrangements. Project C, the only collaboration involving formal civil society inclusion, revealed tensions between symbolic inclusion and substantive participation. Although civil society actors were formally represented, participants described progressive

marginalisation from agenda-setting and strategic decision-making processes. This suggests that formally including a fourth-helix actor does not automatically yield the democratic and inclusive collaboration envisioned in Quadruple Helix theory (Carayannis & Campbell, 2009; Carayannis et al., 2022). Rather, our findings point toward a potentially important distinction between formal inclusion and substantive participatory authority. Without governance arrangements that actively institutionalise meaningful participation, Quadruple Helix collaborations may reproduce underlying Triple Helix power asymmetries within a more complex stakeholder environment.

This distinction between formal inclusion and substantive participatory authority has broader theoretical implications for how Quadruple Helix arrangements are designed and evaluated. Carayannis and Campbell (2009) envision the fourth helix as introducing knowledge democracy and civic epistemology into innovation systems, implying not merely the presence of civil society actors but their genuine influence over agenda-setting and strategic direction. Our findings suggest that without explicit governance mechanisms protecting this influence, such as co-decision rights, joint priority-setting processes, and recognition of community accountability logics as legitimate alongside academic and commercial ones, civil society inclusion risks becoming a legitimising veneer over what remains a Triple Helix power structure. This has important implications for how funders and policy-makers design and evaluate Quadruple Helix initiatives: formal composition of the partnership is an insufficient criterion for assessing whether QH principles are being realised in practice. Evaluation frameworks attentive to participation quality, influence distribution, and accountability logic negotiation would be more diagnostically useful than partner counts alone.

A cross-cutting observation concerns the specific role of higher education institutions in the failure dynamics we document. While universities were formally present as partners in all three cases, our findings suggest that their institutional characteristics actively contributed to failure rather than serving as neutral backdrops. The publication-based performance evaluation systems documented in Section 4.3 created a structural disincentive to sustain collaboration. As workloads increased, academic participants faced a genuine career logic that deprioritised partnership activity, which was unrewarded by their home institutions. University governance structures also contributed to key-person dependency in Projects A and B, where continuity of collaboration depended on individual champions operating outside formal institutional mandates rather than on institutionalised university-level commitments. In Project C, the university's orientation toward academically legitimate outputs created a logic conflict with both the civil society intermediary and the

regional government agency that was never explicitly negotiated. These observations suggest that Helix collaboration research would benefit from treating universities not merely as one of three required actor types but as institutionally complex organisations whose internal incentive structures, governance arrangements, and academic logic actively shape whether and how partnerships survive.

Fifth, existing accounts of collaboration decline focus primarily on structural and relational factors while paying limited attention to the cognitive dimension of how participants process and respond to warning signs. Drawing on Weick's (1995) account of sensemaking under commitment and the optimism bias literature (Sharot, 2011), our findings point toward a cognitive mechanism, which we term collaborative optimism bias, that may help explain why structural and relational deterioration so consistently goes unaddressed until recovery becomes difficult. This refers to the inclination of invested participants to interpret ambiguous signals of decline as temporary fluctuations rather than as indicators of deeper structural deterioration, a tendency motivated by the reputational, relational, and sunk-cost stakes of acknowledging failure.

This mechanism is consistent with Staw's (1981) account of escalation of commitment, in which decision-makers increase investment in a failing course of action partly to justify prior commitments, and with Brockner's (1992) analysis of the psychological entrapment dynamics that discourage exit from deteriorating situations. In collaboration contexts, these dynamics are amplified by the multi-party nature of the relationship: acknowledging failure requires not only personal admission of a poor investment but the public erosion of a shared project identity that may have sustained participants' institutional reputations and relationships.

In the context of collaboration, this cognitive tendency has important organisational implications. Across all three cases, participants retrospectively recognised warning signs only after deterioration had progressed to a point where recovery became increasingly difficult. Conceptually, collaborative optimism bias may therefore function as a mechanism that amplifies structural and relational vulnerabilities by delaying corrective intervention. While exploratory, this interpretation suggests that future research on collaboration resilience may benefit from greater attention to sensemaking processes, escalation dynamics, and the organisational conditions under which warning signs become normalised or discounted.

5.2 *Practical Implications*

Our findings offer several practical implications for those designing, managing, or funding university–stakeholder collaborations.

Use the failure typology diagnostically: The five failure types- dissolution, zombie collaboration, stillbirth, fragmentation, and mission drift- can serve as a diagnostic framework. Practitioners should periodically assess which trajectory their collaboration is following. The zombie state deserves particular attention: formal existence without meaningful activity may be the most common and least-recognised form of failure, and it is often easier to prevent than to reverse once established.

Invest in governance before the need: The consistent finding that structural vulnerability preceded and enabled failure suggests that governance investment should be front-loaded. Clear roles, transparent decision-making, conflict-resolution mechanisms, and documented succession plans are not bureaucratic overhead: they are the infrastructure that enables collaborations to absorb the inevitable shocks and transitions that occur over multi-year timescales (Provan & Kenis, 2008; Huxham & Vangen, 2005).

Negotiate success explicitly in formation: Institutional misalignment was latent in all three cases, becoming visible only under operational pressure. Explicitly negotiating what 'success' means for each partner (including timelines, output types, and accountability structures) at the formation stage can surface these tensions when they are still manageable.

Plan sustainability from the beginning: In a Finnish context marked by EU project funding cycles, the transition from funded to self-sustaining operation is the most predictable crisis point. Sustainability planning should begin at project initiation, not when funding is running out.

Monitor relational health as a leading indicator: Our findings suggest that relational erosion, declining meeting attendance, slower communication, and reduced information sharing precede structural and environmental crises. Simple monitoring mechanisms that track participation quality, not just activity volume, can provide early warning signals that enable intervention before cascading failure begins.

Take Quadruple Helix inclusion seriously: For collaborations involving civil society partners, formal inclusion is not sufficient. Governance mechanisms that protect meaningful participation, co-decision rights, joint priority-setting, and explicit recognition of different accountability logics are necessary to realise the legitimacy and democratic benefits that Quadruple Helix arrangements promise. This is consistent with Carayannis and Campbell's (2009) original normative vision of the fourth helix as a site of knowledge democracy rather than symbolic representation.

6 Conclusion

University–stakeholder collaboration has become a central strategy for regional innovation policy, yet our understanding of such collaborations remains profoundly incomplete because we have studied primarily those that succeeded. This study addressed that gap by examining why three university-stakeholder collaborations in Finland collapsed and by developing and applying a conceptual framework that distinguishes failure types, stages, and contributing factors.

Our findings reveal that collaboration failure is neither simple nor monocausal. It emerges from cascading interactions among relational, structural, institutional, and environmental factors, often following a recognisable process: a period of gradual, misread decline; a critical juncture that exposes latent vulnerabilities; and cascading breakdown that ends not in formal closure but in informal dissolution. Structural vulnerability, stemming from inadequate governance and key-person dependency, was a prominent and recurring condition across all cases. At the same time, environmental triggers served as proximate causes, revealing pre-existing weaknesses. For Helix theory, these findings show that the presence of university, industry, and government actors is necessary but not sufficient: the transition from interpersonal relationships to stable governance structures is a critical passage that many collaborations fail to navigate. For Quadruple Helix theory specifically, formal inclusion of civil society does not guarantee the substantive participation that distinguishes genuine QH collaboration from renamed Triple Helix arrangements.

Several limitations should be noted when interpreting these findings. The study examined only failed collaborations, meaning we cannot definitively distinguish conditions associated with failure from those that also appear in successful cases. Our reliance on retrospective accounts introduces potential memory and attribution biases, mitigated but not eliminated through triangulation with documentary evidence. The Finnish context, with its well-developed university-government relations, EU-funded regional development programmes, and strong publication-based academic incentives, provides a specific backdrop that readers should consider when assessing transferability. Finally, the empirical base of six interviews, while designed for depth rather than breadth, limits the range of perspectives captured, and future studies with larger participant samples would strengthen the empirical foundation.

These limitations point toward several productive directions for future research. Comparative studies examining matched pairs of successful and failed collaborations would enable stronger claims about what distinguishes sustainable from fragile partnerships. Longitudinal studies tracking collaborations

from formation would provide richer evidence of temporal dynamics and earlier identification of warning signs than retrospective accounts allow. Quantitative and intervention-based studies would complement our qualitative depth with systematic breadth, examining failure rates across collaboration types and testing structured early-warning mechanisms. Finally, comparative cross-national studies would assess how different national innovation system characteristics, including funding architecture, university governance models, and civil society traditions, shape the prevalence and form of Helix collaboration failure.

The failed helix deserves as much scholarly attention as the successful one. In learning systematically from failure, we may contribute more to the success of future collaborations than by continuing to catalogue the features of those that happened to survive.

References

Ampuja, M., & Horowitz, M. (2025). Doing "more with less": The entrepreneurialization of Finnish higher education and innovation policy discourses in 2015–2019. Scandinavian Journal of Educational Research, 69(2), 303–317. https://doi.org/10.1080/00313831.2023.2299988.

Ariño, A., & de la Torre, J. (1998). Learning from failure: Towards an evolutionary model of collaborative ventures. Organization Science, 9(3), 306–325. https://doi.org/10.1287/orsc.9.3.306.

Blumer, H. (1954). What is wrong with social theory? American Sociological Review, 19(1), 3–10. https://doi.org/10.2307/2088165.

Brockner, J. (1992). The escalation of commitment to a failing course of action: Toward theoretical progress. Academy of Management Review, 17(1), 39–61. https://doi.org/10.5465/AMR.1992.4279568.

Bryson, J. M., Crosby, B. C., & Stone, M. M. (2015). Designing and implementing cross-sector collaborations: Needed and challenging. Public Administration Review, 75(5), 647–663. https://doi.org/10.1111/puar.12432.

Cai, Y., & Lattu, A. (2022). Triple Helix or Quadruple Helix: Which model of innovation to choose for empirical studies? Minerva, 60, 257–280. https://doi.org/10.1007/s11024-021-09453-6.

Carayannis, E. G., & Campbell, D. F. J. (2009). "Mode 3" and "Quadruple Helix": Toward a 21st century fractal innovation ecosystem. International Journal of Technology Management, 46(3–4), 201–234. https://doi.org/10.1504/IJTM.2009.023374.

Carayannis, E. G., & Campbell, D. F. J. (2010). Triple Helix, Quadruple Helix and Quintuple Helix and how do knowledge, innovation and the environment relate to

each other? International Journal of Social Ecology and Sustainable Development, 1(1), 41–69. https://doi.org/10.4018/jsesd.2010010105.

Carayannis, E. G., Campbell, D. F. J., & Grigoroudis, E. (2022). Helix trilogy: The Triple, Quadruple, and Quintuple Innovation Helices from a theory, policy, and practice set of perspectives. Journal of the Knowledge Economy, 13(3), 2272–2301. https://doi.org/10.1007/s13132-021-00813-x.

Charmaz, K. (2014). Constructing grounded theory (2nd ed.). Sage, ISBN 978-0857029133.

Dana, L. P., & Dumez, H. (2015). Qualitative research revisited: Epistemology of a comprehensive approach. International Journal of Entrepreneurship and Small Business, 26(2), 154–170. https://doi.org/10.1504/IJESB.2015.071822.

Denzin, N. K. (1978). The research act: A theoretical introduction to sociological methods (2nd ed.). McGraw-Hill, ISBN 0070163618.

Eisenhardt, K. M. (1989). Building theories from case study research. Academy of Management Review, 14(4), 532–550. https://doi.org/10.5465/amr.1989.4308385.

Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. Academy of Management Journal, 50(1), 25–32. https://doi.org/10.5465/amj.2007.24160888.

Etzkowitz, H., & Leydesdorff, L. (1995). The Triple Helix – university–industry–government relations: A laboratory for knowledge-based economic development. EASST Review, 14(1), 14–19. Disponível em: https://ssrn.com/abstract=248008.

Etzkowitz, H., & Leydesdorff, L. (2000). The dynamics of innovation: From national systems and "Mode 2" to a Triple Helix of university–industry–government relations. Research Policy, 29(2), 109–123. https://doi.org/10.1016/S0048-7333(99)00055-4.

European Commission Policy Support Facility. (2025). Country review to improve research–business collaboration in Finland. European Commission. Available at: https://projects.research-and-innovation.ec.europa.eu/en/statistics/policy-support-facility/policy-support-facility-concludes-country-review-improve-research-business-collaboration-finland.

Gehman, J., Glaser, V. L., Eisenhardt, K. M., Gioia, D., Langley, A., & Corley, K. G. (2018). Finding theory–method fit: A comparison of three qualitative approaches to theory building. Journal of Management Inquiry, 27(3), 284–300. https://doi.org/10.1177/1056492617706029.

Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. Organizational Research Methods, 16(1), 15–31. https://doi.org/10.1177/1094428112452151.

Gray, D., Sundstrom, E., Tornatzky, L. G., & McGowen, L. (2011). When Triple Helix unravels: A multi-case analysis of failures in industry–university cooperative research centres. Industry and Higher Education, 25(5), 333–345. https://doi.org/10.5367/ihe.2011.0057.

Guba, E. G., & Lincoln, Y. S. (1994). Competing paradigms in qualitative research. In N. K. Denzin & Y. S. Lincoln (Eds.), Handbook of qualitative research (pp. 105–117). Sage, ISBN 0803946791.

Guest, G., Bunce, A., & Johnson, L. (2006). How many interviews are enough? An experiment with data saturation and variability. Field Methods, 18(1), 59–82. https://doi.org/10.1177/1525822X05279.

Huxham, C., & Vangen, S. (2005). Managing to collaborate: The theory and practice of collaborative advantage. Routledge. https://doi.org/10.4324/9780203015407.

Kaiser, K. (2009). Protecting respondent confidentiality in qualitative research. Qualitative Health Research, 19(11), 1632–1641. https://doi.org/10.1177/1049732309350879.

Mellahi, K., & Wilkinson, A. (2004). Organizational failure: A critique of recent research and a proposed integrative framework. International Journal of Management Reviews, 5–6(1), 21–41. https://doi.org/10.1111/j.1460-8545.2004.00095.x.

Patton, M. Q. (2015). Qualitative research & evaluation methods: Integrating theory and practice. Sage, ISBN 978-1412972123.

Provan, K. G., & Kenis, P. (2008). Modes of network governance: Structure, management, and effectiveness. Journal of Public Administration Research and Theory, 18(2), 229–252. https://doi.org/10.1093/jopart/mum015.

Research Council of Finland. (2025). Data dashboards. Disponível em: https://www.aka.fi/en/from-research-to-society/rcf-impact/data-dashboards/.

Ring, P. S., & Van de Ven, A. H. (1994). Developmental processes of cooperative interorganizational relationships. Academy of Management Review, 19(1), 90–118. https://doi.org/10.5465/AMR.1994.9410122009.

Sharot, T. (2011). The optimism bias. Current Biology, 21(23), R941–R945. https://doi.org/10.1016/j.cub.2011.10.030.

Staw, B. M. (1981). The escalation of commitment to a course of action. Academy of Management Review, 6(4), 577–587. https://doi.org/10.5465/amr.1981.4285694.

Thomasson, A., Wigren-Kristoferson, C., & Scheller, C. (2021). What constitutes failure? The influence of public interests in securing accountability in Triple Helix initiatives. Triple Helix, 8(1), 128–162. https://doi.org/10.1163/21971927-bja10020.

Thornton, P. H., Ocasio, W., & Lounsbury, M. (2012). The institutional logics perspective: A new approach to culture, structure and process. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199601936.001.0001.

Timmermans, S., & Tavory, I. (2012). Theory construction in qualitative research: From grounded theory to abductive analysis. Sociological Theory, 30(3), 167–186. https://doi.org/10.1177/0735275112457914.

Vangen, S., & Huxham, C. (2003). Nurturing collaborative relations: Building trust in interorganizational collaboration. Journal of Applied Behavioral Science, 39(1), 5–31. https://doi.org/10.1177/0021886303039001001.

Weick, K. E. (1995). Sensemaking in organizations. Sage, ISBN 978-0803971776.
Williams, P. (2002). The competent boundary spanner. Public Administration, 80(1), 103–124. https://doi.org/10.1111/1467-9299.00296.
Yin, R. K. (2018). Case study research and applications: Design and methods (6th ed.). Sage, ISBN 978-1506336169.